R for Earth Data: Quick-Start Cheat Sheet

The first commands from R for Earth and Sustainability Data Analytics

Learn R · Explore Earth

1. Get set up

Install R from r-project.org, then RStudio from posit.co. Both are free. Always open RStudio, not R directly.

Write your code in the Script (top left). Results appear in the Console (bottom left).

Run a line: place the cursor on it and press Ctrl + Enter (Command + Enter on Mac). Save the script with Ctrl + S.

2. Your first commands

```
5 + 3        # arithmetic
1:10         # a sequence 1,2,...,10
x <- 10      # store 10 in object x
x            # show what is in x
```

The arrow <- means "put this value into this object". The # symbol starts a comment; R ignores the rest of the line.

3. Vectors and a first plot

```
day <- 1:5
rainfall_mm <- c(0, 12, 5, 0, 8)
plot(day, rainfall_mm)
```

c() combines values into a vector (a group of values of one type). Keep a vector all numbers or all text: "5" in quotes is text, not a number you can calculate with.

4. Look at your data

```
d <- read.csv("mydata.csv")  # read a table
head(d)     # first few rows
str(d)      # columns and their types
summary(d)  # quick statistics
names(d)    # column names
nrow(d)     # number of rows
```

Use d$Temp to pick one column, then mean(d$Temp) for its average or sd(d$Temp) for its spread.

5. Missing values and dates

Real Earth data often has gaps and dates.

```
is.na(d$Temp)        # TRUE where missing
d2 <- na.omit(d)     # drop missing rows
d$date <- as.Date(d$DATE)  # text to date
```

6. Packages

```
install.packages("sf")  # once per computer
library(sf)             # each new session
```

install.packages downloads a package; library loads it so you can use its tools.

7. Getting help, and when R gets stuck

Type ?mean to open the help page for a function.

If the Console shows a + and waits, a bracket or quote is unfinished. Click the Console and press Esc to start over.

Read error messages slowly. Errors are a normal part of learning.